



National Archives and Records Administration

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LIST OF DOCUMENTATION

System and/or File Title:

Cambodian Incidents (KHMER) File;
July, 1970 –May, 1974

	Number of Pages
I. NARA Documentation	
1. List of Documentation (Folder 1)	1 pg.
2. User Note (Folder 1)	13 pg.
3. Translation File Record Layout (Folder 2)	1 pg.
4. Translation Programs (Folders 3-5)	16 pg.
II. Agency Documentation	
1. File Description, File Purpose, KHMER Data Element Definitions, KHMER File Format Table, KHMER Input Sources, Update Procedures, and Code Tables (Folder 6)	49 pg.
2. File Description, File Purpose, KHMER Data Element Definitions, KHMER File Format Table, KHMER Input Sources, Update Procedures, and Code Tables (Copy) (Folder 7)	49 pg.
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III. NARA Processing Materials	
1. Manual Verification of NIPS File (Folder 8)	
2. Automated Validation of Electronic Records (Folder 9)	
3. File Sample Printouts (Folders 10-15)	

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Prepared by: J. Powell, Archives Specialist

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**USER NOTE**

System and/or File Title: **Cambodian Incidents (KHMER File);
July, 1970 – May, 1974**

I. National Military Command System Information Processing System (NIPS)

Overview. The National Military Command System Information Processing System 360 Formatted File System, commonly referred to as NIPS or NIPS 360 FFS, was developed in the 1960's under contract with the International Business Machines Corporation (IBM). It was a data management system which was operational on IBM System / 360 and System / 370 computers. NIPS provided data management support to structure files, generate and maintain files, revise and update files and data, select and retrieve data, and generate reports. NIPS used an IBM generalized file management system known at IBM as the Formatted File System (FFS). Originally implemented on the IBM 1410 computer, NIPS was primarily run on the IBM System 360 and was used for a variety of data related to the Vietnam War. The Cambodian Incidents (KHMER) File; July, 1970 – May, 1974 was transferred to the National Archives and Records Service (NARS) in the NIPS format.

NIPS Format. Each NIPS file is organized into variable-length records, blocked and spanned. Although the bulk of the file is data, the beginning of the file consists of supporting information used during file maintenance, data retrieval, and output processing. The beginning of the file contains information such as the security classification record, data file control records, element format (field definitions / File Format Table) records, and file maintenance logic statement records. The sixth character of each logical record (following the four-character logical record length indicator and a system character) in the data file is used as a code indicating the type of information in that record (B = classification record, C = data file control record, F = element format record, L = file maintenance logic statement record, and R = data records).

NIPS Documentation. A single set of the NIPS documentation is located in Stack 530 in the documentation for accession NN3-218-76-025, Combat Air Summary File (OPREA), 1962-73 (AKA Air Operations File). The table below lists the titles of the NIPS documentation.

Title	Date
National Military Command System Information Processing System 360 Formatted File System (NIPS 360 FFS), User's Manual, Volume I, Introduction to File Concepts	1-Jul-71
National Military Command System Information Processing System 360 Formatted File System (NIPS 360 FFS), User's Manual, Volume II, File Structuring (FS)	1-Jul-71
National Military Command System Information Processing System 360 Formatted File System (NIPS 360 FFS), User's Manual, Volume III, File Maintenance (FM)	1-Jul-71

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Title	Date
National Military Command System Information Processing System 360 Formatted File System (NIPS 360 FFS), User's Manual, Volume V, Output Processor (OP)	1-Jul-71
National Military Command System Information Processing System 360 Formatted File System (NIPS 360 FFS), User's Manual, Volume VI, Terminal Processing (TP)	1-Jul-71
National Military Command System Information Processing System 360 Formatted File System (NIPS 360 FFS), User's Manual, Volume VII, Utility Support (UT)	1-Jul-71
National Military Command System Information Processing System 360 Formatted File System (NIPS 360 FFS), User's Manual, Volume VIII, Job Preparation Manual	1-Jul-71
National Military Command System Information Processing System 360 Formatted File System (NIPS 360 FFS), User's Manual, Volume IX, Error Codes	1-Jul-71
National Military Command System Information Processing System 360 Formatted File System (NIPS 360 FFS), Technical Report TR 80-72, NIPS Processing Handbook	1-Feb-73
NMCS Information Processing System 360 Formatted File System (NIPS 360 FFS), General Description	1-Sep-78
NMCS Information Processing System 360 Formatted File System (NIPS 360 FFS), User Manual, Volume V, Output Processor (OP)	1-Sep-78
NMCS Information Processing System 360 Formatted File System (NIPS 360 FFS), User Manual, Volume VI, Terminal Processing (TP)	1-Sep-78
NMCS Information Processing System 360 Formatted File System (NIPS 360 FFS), User Manual, Volume VII, Utility Support	1-Sep-78
NMCS Information Processing System 360 Formatted File System (NIPS 360 FFS), User Manual, Volume VIII (Revised) Job Preparation	1-Sep-78
NMCS Information Processing System 360 Formatted File System (NIPS 360 FFS), User Manual, Volume, IV, Retrieval & Sort Processor (RASP)	1-Sep-78
NMCS Information Processing System 360 Formatted File System (NIPS 360 FFS), Users Manual, Volume I, Introduction to File Concepts	1-Sep-78
NMCS Information Processing System 360 Formatted File System (NIPS 360 FFS), Users Manual, Volume II, File Structuring (FS)	1-Sep-78
NMCS Information Processing System 360 Formatted File System (NIPS 360 FFS), Users Manual, Volume III, File Maintenance (FM)	1-Sep-78
NMCS Information Processing System 360 Formatted File System (NIPS 360 FFS), Users Manual, Volume IX, Error Codes	1-Sep-78
NMCS Information Processing System 360 Formatted File System (NIPS 360 FFS), Installation of NIPS 360 FFS	1-Sep-78

II. File Translation Process

Background. The Machine-Readable Archives Division (NNR) of NARS received the original Cambodian Incidents (KHMER) File in the NIPS format. Because the NIPS application was not widely available and IBM's support for NIPS was declining, NNR made a decision in the late 1970's that the files should be maintained in a software-independent format. Using the accompanying paper documentation and the File Format Table (FFT) at the beginning of the NIPS data file, an output record layout was created using the NIPS Output Processor (OP). Using the NIPS application, a second data file with a new fixed

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length format was created by NNR. This process of creating the file in the new format as an output from the NIPS application was known within NNR as “de-NIPS-ing” and the resulting data file was known as a “de-NIPS’d” file. The “de-NIPS-ing” process deleted all of the file content information at the beginning of each NIPS file and copied only the actual data records into a fixed-length format. The first fixed-length record of each “de-NIPS’d” record contains the control set, the fixed set, and the first occurrence of each of the periodic sets. Additional “de-NIPS’d” records were written for instances where there were multiple occurrences of the periodic sets. The additional records did not contain the data from the control set or fixed set. The records are linked by the control set which contains one or more fields which as a set are unique for each record. A single “de-NIPS’d” file in an Extended Binary Coded Decimal Interchange Code (EBCDIC) character set was created using the NIPS application. The table below lists the “de-NIPS’d” KHMER file created by NNR listing the XMIS number and file / dataset name. The XMIS number is a unique number assigned to each file.

“De-NIPS” Files	
XMIS #	File Name
005190	KHMER_7001_7404

Vietnam NIPS Files Translation Project Tasking. Three issues were addressed by the Electronic and Special Media Records Services Division (NWME) of the National Archives and Records Administration (NARA) starting in 2002. The first issue was that a number of the NIPS files in different series had not been “de-NIPS’d” in the late 1970’s. The second issue involved the “de-NIPS’d” files containing additional records with the multiple occurrences of the periodic sets. Any re-ordering of the sequence of the records in that “de-NIPS’d” file caused those records without control and fixed set data to lose their linkage or relationship to the first record containing the control and fixed set data. A decision was made by NWME to create a second set of files for selected series with a fixed-length format similar to the “de-NIPS’d” files but with every record containing the control set and fixed set data. Because the NIPS application was no longer available, translation programs would be written using a programming language that would create a file with a fixed length format from the original NIPS file. The third issue was that the files in the NIPS format would be maintained as permanent based upon a decision of the Director, NWME on April 2, 2002.

File Identification. The Cambodian Incidents (KHMER) File was identified using the Archival Preservation System (APS) catalog database. A search query was formulated to provide all catalog records for data files in Record Group 330. KHMER is assigned to Record Group 330. The output report provided information for Record Group 330 that included the following: short title, file name, and accession number. This information is derived from data found in 5 fields in the File Master table in the APS catalog database. The short title is the field “stitle” in the File Master table. The file name is the field “fid” in the File Master table. The accession number is a combination of 3 fields “rgnum,” “accyear,” and “accitem” in the File Master table. The accession number for the KHMER File; July, 1970 – May, is NN3-330-76-039. The KHMER NIPS file had the following data in the previously described fields: an accession number of 330-1976-039, the alphabetic

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character string of "KHMER" and "NIPS" in the short title. Additionally each file has a unique number in the APS catalog database know as an XMIS number which is the field "fxno" in the File Master table. The table below lists the KHMER NIPS file identified from the APS catalog.

XMIS #	Accession Number	Short Title	File Name
015452	330-1976-039	KHMER_7001_7404_NIPS	(blank)

Reference Request. A copy of the KHMER NIPS file was requested as a standard reference request to NWME Reference. NWME Reference forwarded the request to NWME Technical Services who produced the copy of the file. The KHMER NIPS file was requested on compact disk (CD) with a specification that the records be padded to the longest record length. At the time of the reference request, the KHMER NIPS file had not been preservation copied and the file was on the original agency 9-track magnetic tape.

The following is the process used by NWME Technical Services to copy the KHMER NIPS file.

1. An attempt was made to make a copy of the file from the 9-track tape reel to a 3480 magnetic cartridge. This process failed and the 9-track tape reel was determined to be unreadable. A utility copy on a 3480 magnetic cartridge was found which contained a readable KHMER NIPS file.
2. The file on the 3480 magnetic cartridge was tape mapped to determine the longest record length, which was 996 record positions.
3. The APS (version 5.17.8.3) smart copy function was used to re-block and copy the files onto the reference workstation's hard drive. The workstation had Microsoft Windows 2000 as an operating system.
4. According to the APS catalog, the record format should have been variable; however, it was fixed. The input was changed to D/01000/01004. The character set remained the same. Since the file was not named, it was given a file identifier of KHMER7074. The file was copied as fixed blocks, padded to the longest record length (F\996\996\1), without delimiters.
5. APS is unable to compare a variable blocked file (input) to a file copied onto random access memory or hard drive (output) and for that reason the files were not compared.
6. The files were then copied from the work station hard drive to a recordable compact disk (CDR) using Adaptec CD Easy CD Creator (version 4).
7. Using APS again, a directory of the CDR was printed and the file on the CDR was dumped.

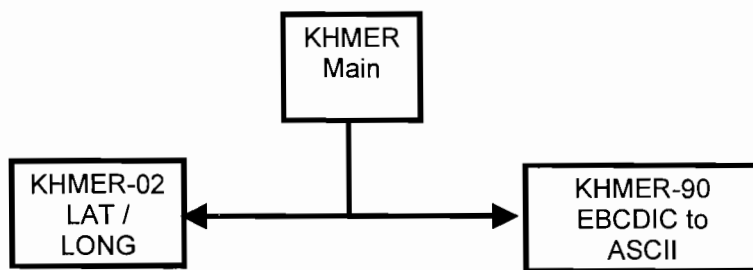
The table below lists the KHMER NIPS file provided in the reference request.

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XMIS #	File Name	Size (bytes)	# of Records	CD External Label	CD Internal Label
015452	KHMER7074	9,163,200	67,661	KHMER7074 Job #2003E02 CDR#2003E01-1	2003e01-1

Translation Programs. The NIPS file was converted into the new fixed length format by using three programs written in Common Business-Oriented Language (COBOL). The COBOL programs were created using MERANT Micro Focus Personal COBOL for Windows 3.1 V 1.1 (PCS). The program KHMER (Folder 3, Main Program Text File) is the main program and it calls the other two programs. KHMER-02 (Folder 4, Latitude Longitude Conversion Program Test File) is the sub-program that translates the Latitude/Longitude field and KHMER-90 (Folder 5, ASCII Conversion Sub-Program Text File) is the sub-program that translates the EBCDIC / binary characters into Extended American Standard Code for Information Interchange (ASCII) characters. The schematic below depicts the translation programs and their relationships.



Record Format.

Because each NIPS 360 FFS system contained records of various sizes, the files all used variable length records in variable length blocks. This was a standard IBM method used to write records on magnetic tape files when not all the records were the same size. In KHMER, there were two different record types (one fixed type and one periodic type) and each record type had its own distinct length.

In NIPS, as in all FFS systems, the meta-data was written onto the magnetic tape. The meta-data contained information about the names of the fields, their size, their location, etc.

The terminology used in this file management system is somewhat unique. Data is organized into "Data Records" each of which contains one "Fixed Set" followed by none or more "Periodic Sets". There is only one type of Periodic Set in KHMER. To illustrate this, an example of four hypothetical Data Records is listed below:

- Data Record 1 = Fixed Set A + Periodic Set #1 A1
- Data Record 2 = Fixed Set B
- Data Record 3 = Fixed Set C + Periodic Set #1 C1 + Periodic Set #1 C2
- Data Record 4 = Fixed Set D + Periodic Set #1 D1

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If the four hypothetical Data Records shown above were the complete NIPS file, then that NIPS file would contain eight records. The first Data Record (A) would have two records, the second Data Record (B) would have only one record, the third Data Record (C) would have three records, and the fourth Data Record (D) would have two records.

For the translated files, each record has a fixed length of 187 record positions. Each record includes the record positions for the fixed set and for the periodic set. The data for the fixed set is always present within each translated record and specific fields within the fixed set known as the control set serve as the unique identifier for each record. The first instances of the periodic set are found in the translated record. Additional instances of the periodic set data are written to additional records but with the same fixed set data as the first instance. Using the hypothetical NIPS file shown above, the eight NIPS records would become five translated records. The new arrangement would be:

Translated Record 1 = Fixed Set A + Periodic Set #1 A1
Translated Record 2 = Fixed Set B
Translated Record 3 = Fixed Set C + Periodic Set #1 C1
Translated Record 4 = Fixed Set C + Periodic Set #1 C2
Translated Record 5 = Fixed Set D + Periodic Set #1 D1

Record Layout.

NWME created a record layout for the translation data file using the element format (field definitions / File Format Table) in the KHMER NIPS file and the agency documentation. The record layout (Folder 2, Translation File Record Layout) contains the Column Name, Title / Description, Data Type, Start Position, End Position, and Column Width. The last column in the record layout text file named "LINE-END" was not derived from the NIPS file or agency documentation and was added during the translation process. The "LINE-END" column accounts for two characters that were inserted at the end of each record (Record Positions 188 – 189) in the translation data file when it was originally copied to CDR before preservation copying. The two characters, indicating the end of a record, were not transferred to the preservation copy of the translation data file. The record length of the translation data file is a total of 187 record positions as indicated in the element format (field definitions / File Format Table) of the NIPS file and the agency documentation. The table below depicts the record layout of the KHMER translation data file. The location of the columns in the control set, fixed set, and periodic set are indicated on the left.

Set	Column Name	Title	Data Type	Start Position	End Position	Width
Control	MSGDATE	Message Date	VARCHAR2	1	6	6
	ITEMNO	Item Number	VARCHAR2	7	8	2
Fixed	FLG1	ARVN Flag	VARCHAR2	9	9	1
	FLG2	Second Flag	VARCHAR2	10	10	1
	FLG3_FLG4	Third & Fourth Flags	VARCHAR2	11	18	8
	MRSD	Military Region Subdivision	VARCHAR2	19	22	4

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Set	Column Name	Title	Data Type	Start Position	End Position	Width
	ADCT	Conversion Code	VARCHAR2	23	23	1
	ADCR	Military / Political Subdivision Code	VARCHAR2	24	26	3
	INCDATE	Incident Date	VARCHAR2	27	32	6
	QTR	Quarter	VARCHAR2	33	36	4
	HY	Half Year	VARCHAR2	37	40	4
	INIT	Initiator	VARCHAR2	41	41	1
	EF	Initiator Code	VARCHAR2	42	42	1
	IACT	Initiator Activity	VARCHAR2	43	44	2
	OBJECT	Objective	VARCHAR2	45	46	2
	RACT	Recipient Activity	VARCHAR2	47	48	2
	EWEAP	Enemy Weapon	VARCHAR2	49	50	2
	FACT	Friendly Activity	VARCHAR2	51	52	2
	EACT	Enemy Activity	VARCHAR2	53	54	2
	UTM	Universal Transverse Mercator	VARCHAR2	55	62	8
	CORD	Coordinate	VARCHAR2	63	77	15
	LATLONG	Latitude / Longitude	VARCHAR2	78	92	15
	ITIME	Incident Time	VARCHAR2	93	96	4
	WK	Week	VARCHAR2	97	100	4
	FNUMB	Friendly Unit Number	VARCHAR2	101	103	3
	FTYPE	[Friendly] Unit Type	VARCHAR2	104	104	1
	FSIZE	[Friendly Unit] Size	VARCHAR2	105	105	1
	FNAT	Friendly [Unit] Nationality	VARCHAR2	106	106	1
	ESIZE	Enemy [Unit] Size	VARCHAR2	107	107	1
	ENAT	Enemy [Unit] Nationality	VARCHAR2	108	108	1
	LOCNM	Location Name	VARCHAR2	109	123	15
	COMMENT	Comment	VARCHAR2	124	135	12
Periodic	SETCD	Casualty Code	VARCHAR2	136	136	1
	SETEF	Casualty Code	VARCHAR2	137	137	1
	ASSOC	Casualty Nationality	VARCHAR2	138	143	6
	KIA	Killed	NUMBER	144	146	3
	WIA	Wounded	NUMBER	147	149	3
	CAP	Captured / Missing	NUMBER	150	152	3
	RAL	Rallied	NUMBER	153	155	3
	PERCOM	Periodic Comments	VARCHAR2	156	187	32

Original Data Editing and Verification.

During the Vietnam conflict subject matter specialists transcribed the KHMER data onto keypunch coding sheets. Then keypunch operators used these coding sheets to produce the KHMER punch cards. The computer used the punch cards, together with the FFS programs to create the KHMER file for NIPS.

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Geographic Coordinates.

aabbbbbbbbbbccccccccc cccccccc cccccccc deffffff ffgggggggg gggggggggg gggggggggg

bbbbbbbb – An 8-bit unsigned binary number indicating the degrees of latitude measured from the south pole. If the binary number is greater than ninety, then ninety is subtracted from the binary number and the remainder is the latitude in the Northern Hemisphere. If the binary number is less than ninety, then the binary number is subtracted from ninety and the remainder is the latitude in the Southern Hemisphere. If the binary number is equal to ninety, then the latitude is zero (the equator).

d – This is the longitude hemisphere flag.

ffffff – The longitude measured in degrees from 0 degrees (Greenwich). This is represented as an 8-bit unsigned binary number.

Latitude/Longitude on the translated files is shown as **ddmmssxDDDMMSSX** in which:

dd is the 2-digit degrees of Latitude

mm is the minutes of Latitude

ss is the seconds of Latitude

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x equals either North or South

DDD is the 3-digit degrees of Longitude

MM is the minutes of Longitude

SS is the seconds of Longitude

X equals either East or West.

The example of the latitude of 495010N would mean that 49 is the degrees of latitude, 50 is the minutes, 10 is the seconds, and N is North.

Similarly, the longitude of 1032504E would mean 103 degrees, 25 minutes, 4 seconds, East.

Non-Graphic Characters. Each non-graphic character found in the KHMER NIPS file was changed to the caret (^) graphic character in the translated data file. The table below lists the non-graphic characters changed to an ASCII caret (^).

NIPS File		Translated File	
Hexadecimal	EBCDIC	Hexadecimal	ASCII
00	Null	5E	Caret (^)
37	End of Transmission	5E	Caret (^)
C0	Opening Brace ({)	5E	Caret (^)
D0	Closing Brace (})	5E	Caret (^)

ASCII Conversion. Some changes were made to the translated files. The original KHMER NIPS file was coded in the 8-bit Extended Binary Coded Decimal Interchange Code (EBCDIC) and the translated files are all coded in the 8-bit American National Standard Code for Information Interchange – Extended (Extended ASCII). The contents of the binary fields in the NIPS files were changed to ASCII numbers in the translated files.

Translation Data and Documentation File Production. The KHMER translation data file was created by running the translation program with the KHMER NIPS file. The translation programs and the record layout were saved as text files. The translation data file, the three translation programs and the record layout were copied to a single CDR. The table below lists the files which were copied to the CDR.

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Item	File Name	Byte Size	# of Records	CD External Label	CD Internal Label
Translated Data File	KHMER7074	9,163,200	50,008	KHMER CD#2 1 Translation File 1 Record Layout File 3 Program Files	KHMER02
KHMER (Main Program)	KHMER.CBL	13,746			
KHMER 02 (Lat / Long Program)	Khmer-02.cbl	19,569			
KHMER 90 (ASCII Program)	KHMER-90.CBL	5,786			
Record Layout	KHMER Record Layout.txt	1,461			

Initial Verification. The translated data file was reviewed in Microsoft Excel to determine if the translated data matched the record layout. The translated data file was then verified on the Archival Electronic Records Inspection and Control (AERIC) system before being submitted for preservation copying.

Preservation and Verification. The KHMER NIPS file, the KHMER translation data file, the three translation programs text files, and the record layout text file were submitted for preservation copy using the established NWME procedures. The table below lists the resulting preservation copies of the six files.

Item	XMIS #	File Name	Size (bytes)	# of Records	Job Preservation #
NIPS Data File	015452	KHMER.7007.7404.N	8,403,396	67,661	2004-1210
Translated Data File	082221	KHMER.TR.PRG.7074	9,351,496	50,008	2003-5507
KHMER (Main Program)	082223	KHMER.TR.PRG.MAIN	13,746		2003-5509
KHMER 02 (Lat / Long Program)	082224	KHMER.TR.PRG.GRID	19,569		2003-5510
KHMER 90 (ASCII Program)	082225	KHMER.TR.PRG.CONV	5,786		2003-5511
Record Layout	082222	KHMER.TR.LAY.7074	1,461		2003-5508

The KHMER translation files were verified using the established NWME procedures. The KHMER NIPS file was manually verified and was not verified in AERIC because of the file structure previously described.

III. Agency Documentation. Within the portion of the agency documentation titled "Data Element Definitions – KHMER" (Folders 6 or 7, Agency Documentation), for the two fields named FLG3 and FLG4, the FLG3 field is listed as having 3 record positions and the FLG4 field is listed as having 4 record positions. Within the NIPS file in the element format (field definitions / File Format Table) these fields have a total of 8 record positions instead of a total of 7 record positions. The total of 8 record positions is reflected in the translated file record layout.

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IV. Date Ranges of Records in the File. The Message Date (MSGDATE) with the description in the agency documentation of "Date of Message" has records with dates that range from July 1, 1970 – May 26, 1974. The incident Date (INCDATE) with the description in the agency documentation of "Date of Incident" has records with dates that range from June 30, 1970 – June 12, 1974.

V. Data and Agency Documentation.

Minor discrepancies between the data and documentation were discovered during the data verification process. The discrepancies are characters within the data that were not within the expected values listed in the documentation. The explanations of the differences are listed by Data Field Name, in record position order, and are described below.

- ITEMNO: The agency documentation lists a "sequential incident number within a message" in Record Positions 7 – 8. There is a single record for the following values: A1, A2, A3, A4, A5, A6, A7, A8, A9, B0, B1, B2, B3, B4, B5, B6, B7, B8, B9, C1, Z3, Z4, Z5, Z6, Z7, Z8, and Z9.
- MRSD: The agency documentation lists 21 codes in Table 2 – KMRSD (AERIC Code List: KHMER_MIL_SUBDIV) in Record Positions 19 - 22. Some records include the values of 1, 2, 4, 5, 7, 8, 22, 25, 1BAP, 1SRP, 2BAT, 2KIR, 2SRG, 3KCG, 3SMY, 3TPK, 4 KTN, 6MSZ, 7MAZ, and 7MSZ.
- ADCT: The agency documentation lists a value of 7 in Record Positions 23. A large number of records have a value of an asterisk (*) or null (blank).
- ADCR: The agency documentation lists 21 codes in Table 1 – KSD (AERIC Code List: KHMER_POL_SUBDIV) in Record Positions 24 - 26. A larger number of records include the value of three asterisks (***).
- INCDATE: The agency documentation lists a date format of "IY, IM, ID" in Record Positions 27 – 32. A number of records have the values of 7301, 7305, 7306, 7308, 731 1, 7310, 73100, 73102, 73103, 73110, 7312, 73120, 73121, 740, 7401, 74011, 74012, 7402, 74020, 74032, 74041, K30304, K30306, and K30307.
- QTR: The agency documentation lists a date format of "YY'Q'Q" in Record Positions 33 – 36. A small number of records have the values of 73Q and K3Q1.
- HY: The agency documentation lists a date format of "YY'H'H" in Record Positions 37 – 40. A small number of records have the values of 73H and K3H1.
- INIT: The agency documentation lists 14 codes in Table 3 – KNAT (AERIC Code List: KHMER_NAT) in Record Position 41. A single record each has a value of a closing parenthesis ()) and 3.
- IAC: The agency documentation lists 22 codes in Table 10 – KACTION (AERIC Code List: KHMER_ACTION) in Record Positions 43 - 44. A single record each has a value of E and E9.
- OBJECT: The agency documentation lists a combination of 5 codes from Table 9 – KSIZE (AERIC Code List: KHMER_SIZE) in Record Position 45 and 14

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- codes from Table 3 – KNAT (AERIC Code List: KHMER_NAT) in Record Position 46 or the values of 00, 10, 40, 60, and 70 in Record Positions 45 – 46. A small number of records have the values of 1/, 11, or 18.
- RACT: The agency documentation lists 22 codes in Table 10 – KACTION (AERIC Code List: KHMER_ACTION) in Record Positions 47 - 48. A single record each has a value of E and E9.
- EWEAP: The agency documentation lists 17 codes in Table 6 – KWEAP (AERIC Code List: KHMER_WEAP) in Record Positions 49-50. A number of records have the values of 9, 2, 30, 4A, A7, AG, AW, B2, BB, BT, H2, H8, HA, HC, HS, HW, KA, KK, MA, MI, R4, R5, R6, RK, TK, and VV.
- ITIME: The agency documentation lists a value of 0000 – 2400 in Record Positions 93-96. A small number of records have the value of 6, 00, 730, /KO/, 0, 020, 040, 063, 080, 115, 130, 140, 15//, 200, 213, 2405, 2406, 2410, 2415, 2420, 2430, 2440, 2450, 2500, 2550, 2630, 2746, 2818, 2830, 2900, 3000, 3100, 3200, 3500, 3550, 400, 4306, 4419, 4500, 4630, 5000, 5200, 5400, 7000, 8001, 8200, 830, 8500, 9000, and U.O/.
- WK: The agency documentation lists a date format of YYWW in Record Positions 97 – 100. A small number of records have the value of 73 and K3.
- FNUMB: The agency documentation lists the values as “Friendly unit number” in Record Positions 101 – 103. A number of records have the value of /JO, 0/U, 04., 1 2, 15O, 1N, 2F, 7I2, 8AR, 9N, 9Y, AHB, AMB, GB, MHB, PAR, SHB, SHO, SIB, and THB.
- EACT: The agency documentation lists 9 codes of A, C, E, G, I, M, N, P, or T in Record Position 104. A small number of records have the value of o, 2, F, L, R, and S.
- FSIZE: The agency documentation lists 5 codes in Table 9 – KSIZE (AERIC Code List: KHMER_SIZE) in Record Position 105. A small number of records have the values of an opening parenthesis ((), 0, B, F, I, P, and Z.
- FNAT: The agency documentation lists five codes F, J, M, R, and T from Table 3 – KNET (AERIC Code List: KHMER_NAT) in Record Position 106. A small number of records have the value of a semi-colon (;), 0, 4, 6, A, B, G, or I.
- ESIZE: The agency documentation lists 5 codes in Table 9 – KSIZE (AERIC Code List: KHMER_SIZE) in Record Position 107. A small number of records have the values of 6, E, F, H, I, K, N, S, T, U, and Y.
- ENAT: The agency documentation lists five codes E, K, N, and Y from Table 3 – KNET (AERIC Code List: KHMER_NAT) in Record Position 108. A small number of records have the value of a forward slash (/), 0, 1, 2, 9, G, or I.
- LOCNM: The agency documentation lists the “place name as reported in message” in Record Positions 109 – 123. A number of records contain single record position alphanumeric values and textual phrases not associated with a geographical place name.

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 Date: January 26, 2004
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NARA's web site is <http://www.nara.gov>

ASSOC: The agency documentation lists a "six-character name for the nationality / association of the persons suffering casualties" in Record Positions 138 – 143. A number of records contain numeric values of 0, 00, 0000, 000000, 001, 002, 003, 012, 0002, 0004, 1, 1001, 1006, 2, 2001, 2006, 3, 3006, 4, 4011, 4020, 5, 5003, 6, 7, 7008, 8, 8006, 8010, 9, and 9006.

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CAMBODIAN INCIDENTS (KHMER) FILE, RECORD LAYOUT

COLUMN NAME	TITLE / DESCRIPTION	DATA TYPE	START	END	WIDTH
MSGDATE	MESSAGE DATE	DATE	1	6	6
ITEMNO	ITEM NUMBER	VARCHAR	7	8	2
FLAG	FLAG	VARCHAR	9	18	10
MRSD	MILITARY REGION SUBDIVISION	VARCHAR	19	22	4
ADCR	CONVERSION CODE	VARCHAR	23	26	4
INCDATE	INCIDENT DATE	DATE	27	32	6
QTR	QUARTER	VARCHAR	33	36	4
HY	HALF YEAR	VARCHAR	37	40	4
ACTION	ACTION GROUP	VARCHAR	41	48	8
EWEAP	ENEMY WEAPON	VARCHAR	49	50	2
FEACT	FRIENDLY / ENEMY FORCES	VARCHAR	51	54	4
UTM	UNIVERSAL TRANVERSE MERCATOR	VARCHAR	55	62	8
CORD	COORDINATE	VARCHAR	63	77	15
LATLONG	LATITUDE / LONGITUDE	VARCHAR	78	92	15
ITIME	INCIDENT TIME	VARCHAR	93	96	4
WK	WEEK	VARCHAR	97	100	4
FNUMB	FRIENDLY UNIT	VARCHAR	101	103	3
FTYPE	UNIT TYPE	VARCHAR	104	104	1
FSIZE	SIZE	VARCHAR	105	105	1
FNAT	FRIENDLY NATIONALITY	VARCHAR	106	106	1
EUNIT	ENEMY UNIT	VARCHAR	107	108	2
LOCNM	LOCATION NAME	VARCHAR	109	123	15
COMMENT	COMMENT	VARCHAR	124	135	12
SETCD	CASUALTY CODE	VARCHAR	136	136	1
SETEF	CASUALTY CODE	VARCHAR	137	137	1
ASSOC	CASUALTY NATIONALITY	VARCHAR	138	143	6
RESULTS	RESULTS	VARCHAR	144	155	12
PERCOM	PERIODIC COMMENTS	VARCHAR	156	187	32
LINE-END	END OF RECORD MARKERS	VARCHAR	188	189	2



National Archives and Records Administration

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College Park, Maryland 20740-6001

NATIONAL ARCHIVES PRODUCED FILE LISTING

The Office of the Secretary of Defense (OSD), Department of Defense, transferred the Cambodian Incidents File (KHMER), July 1970 – May 1974, data file in the National Military Command and Control System Information Processing System (NIPS) format to the National Archives and Records Administration (NARA). The Archival Services staff in the Electronic and Special Media Records Services Division (NWME) translated the data from the NIPS file into a file containing fixed length records. The technical specifications of the data in the translated file were in an ASCII character set and ANSI labels. The Technical Services staff of NWME copied the data file to one 18-track 3480 tape cartridge, 37,871 bpi, with ANSI standard labels, using an ASCII character set. The data file was blocked at 12,155 characters per block. The NARA staff then produced the following printout from the NARA-created copy; this printout is a character representation of the first few records.

DS#1
D/8930
Storage

FILE #1 BLOCK # 0 RECORD # 0

0000 37 30 30 37 30 31 30 31 20 20 20 20 20 20 20 20
0010 20 20 31 50 56 47 37 31 30 32 37 30 30 37 30 31
0020 37 30 51 33 37 30 48 32 45 45 41 32 34 30 55 32
0030 41 4D 55 32 41 32 57 54 33 32 30 34 35 30 31 31
0040 31 35 34 38 4E 31 30 35 31 37 33 36 45 31 31 31
0050 35 34 38 4E 31 30 35 31 37 33 36 45 20 20 20 20
0060 37 30 32 37 20 20 20 20 39 46 39 45 20 20 20 20
0070 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
0080 20 20 20 20 20 20 20 43 46 43 49 56 49 4C 49 30
0090 30 33 30 30 36 30 30 30 30 30 30 20 20 20 20 20
00A0 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
00B0 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20

70070101

1PVG7102700701

70Q370H2EEA240U2

AMU2A2WT32045011

1548N1051736E111

548N1051736E

7027 9F9E

CFCIVILIO

03006000000

FILE #1 BLOCK # 0 RECORD # 1

00BB 37 30 30 37 30 31 30 32 20 20 20 20 20 20 20 20
00CB 20 20 34 53 52 50 37 34 31 32 37 30 30 36 33 30
00DB 37 30 51 32 37 30 48 31 45 45 48 32 30 30 55 32
00EB 41 4D 55 32 48 32 55 56 37 37 30 37 36 30 31 33
00FB 32 31 30 30 4E 31 30 33 35 31 35 31 45 31 33 32
010B 31 30 30 4E 31 30 33 35 31 35 31 45 20 20 20 20
011B 37 30 32 37 20 20 20 20 39 46 39 45 20 20 20 20
012B 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
013B 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
014B 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
015B 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
016B 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20

70070102

4SRP7412700630

70Q270H1EEH200U2

AMU2H2UV77076013

2100N1035151E132

100N1035151E

7027 9F9E

FILE #1 BLOCK # 0 RECORD # 2

0176 37 30 30 37 30 31 30 33 20 20 20 20 20 20 20 20
0186 20 20 34 4B 54 4D 37 34 31 33 37 30 30 36 33 30
0196 37 30 51 32 37 30 48 31 45 45 48 32 30 30 55 32
01A6 20 20 55 32 48 32 56 56 38 38 30 30 35 30 31 32
01B6 34 32 33 37 4E 31 30 34 35 33 32 32 45 31 32 34
01C6 32 33 37 4E 31 30 34 35 33 32 32 45 20 20 20 20
01D6 37 30 32 37 20 20 20 20 39 46 39 45 20 20 20 20
01E6 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
01F6 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
0206 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
0216 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
0226 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20

70070103

4KTM7413700630

70Q270H1EEH200U2

U2H2VV88005012

4237N1045322E124

237N1045322E

7027 9F9E

FILE #1 BLOCK # 0 RECORD # 3

0231 37 30 30 37 30 31 30 34 20 20 20 20 20 20 20 20
0241 20 20 34 4B 54 4D 37 34 31 33 37 30 30 36 33 30
0251 37 30 51 32 37 30 48 31 45 45 41 31 31 30 52 34
0261 20 20 52 34 41 31 57 55 31 32 30 36 32 30 31 32
0271 31 39 31 37 4E 31 30 35 30 36 33 37 45 31 32 31
0281 39 31 37 4E 31 30 35 30 36 33 37 45 20 20 20 20
0291 37 30 32 37 20 20 20 20 39 46 39 45 20 20 20 20

70070104

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R4A1WU12062012

1917N1050637E121

917N1050637E

7027 9F9E

*** APS RECORD DUMP *** Date: 9/04/2003 Time: 14:19

VID: 021962 VALOC: 018930

02A1 20 20 20 20 20 20 20 20 20 20 20 52 54 20 36 20
02B1 20 20 20 20 20 20 20 20 46 46 46 41 4E 4B 20 20 30
02C1 30 30 30 31 32 30 30 30 30 30 30 20 20 20 20 20
02D1 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
02E1 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20

RT 6
FFFANK 0
00012000000

FILE #1 BLOCK # 0 RECORD # 4

02EC 37 30 30 37 30 32 30 31 20 20 20 20 20 20 20 20
02FC 20 20 31 50 56 47 37 31 30 32 37 30 30 37 30 32
030C 37 30 51 33 37 30 48 32 45 45 48 32 30 30 55 32
031C 41 4D 55 32 48 32 57 54 33 32 30 34 35 30 31 31
032C 31 35 34 38 4E 31 30 35 31 37 33 36 45 31 31 31
033C 35 34 38 4E 31 30 35 31 37 33 36 45 20 20 20 20
034C 37 30 32 37 20 20 20 20 39 46 39 45 20 20 20 20
035C 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
036C 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
037C 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
038C 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
039C 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20

70070201
1PVG7102700702
70Q370H2EEH200U2
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1548N1051736E111
548N1051736E
7027 9F9E